

Engineering Report

Third-Party Repository Investigation

LAB-039

Grafana Repository Analysis
Issue Investigation & Implementation Planning

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1 Executive Summary

This report presents the findings from a systematic investigation of the Grafana open-source repository as part of the KDE Laboratory’s third-party repository capability validation. The investigation focused on identifying, analyzing, and planning fixes for unresolved issues in a mature open-source project.

Context

Objectives Achieved

- Repository successfully cloned and bootstrapped
- Architecture thoroughly understood
- Suitable issues selected for investigation
- Root causes identified with supporting evidence
- Implementation plans generated

1.1 Issues Investigated

Two significant issues were selected from the Grafana issue tracker:

Issue	Description
#116074	Alerting: Manually Set labels not present in notification namespace
#13027	Log axis limits “auto” doesn’t work on histogram (7+ years old)

1.2 Key Findings

Key Finding

- Both investigations successfully identified root causes:
- **Issue #116074:** Template data expansion missing rule labels
 - **Issue #13027:** Y-axis scale distribution hardcoded to linear

2 Investigation 1: Alerting Labels Issue

2.1 Issue Overview

Grafana Issue	#116074
Title	Alerting: Manually Set labels not present in namespace
Created	January 9, 2026
Labels	type/bug, area/alerting, area/backend

2.2 Problem Description

When setting a custom label manually in an alert rule (e.g., `environment: production`), this label does not appear in the `$labels` namespace that is interpolated into notification messages.

Context

Expected Behavior: The manually set label should be present in the `$labels` namespace when using template syntax like `{{ $labels }}` in notification messages.

2.3 Code Flow Analysis

The investigation traced the label expansion through the following components:

1. **Alert Evaluation** → `newState()` in `state/state.go`
2. **Label Expansion** → `expandAnnotationsAndLabels()` in `state/cache.go`
3. **Template Expansion** → `Expand()` in `state/template/template.go`
4. **Alert Conversion** → `StateToPostableAlert()` in `state/compat.go`

2.4 Root Cause

Key Finding

Found in `pkg/services/ngalert/state/cache.go` line 154: The template data used for expanding alert rule labels only contains `extraLabels` and `resultLabels`, not the manually set rule labels themselves.

```
// Current code - template data only has extraLabels + resultLabels
```

```
templateData := template.NewData(mergeLabels(extraLabels, resultLabels),  
    ↪ result)  
  
// When expanding rule labels, $labels doesn't include rule.Labels  
labels, _ := expand(ctx, log, alertRule.Title, alertRule.Labels,  
    ↪ templateData, ...)
```

3 Investigation 2: Histogram Log Axis Issue

3.1 Issue Overview

Grafana Issue	#13027
Title	Log axis limits “auto” doesn’t work on histogram
Created	August 24, 2018 (7+ years ago)
Labels	type/bug, area/panel/graph, help wanted

Important

This is one of the oldest open bugs in the Grafana repository, confirmed present across versions 4.6.3 through 6.5.3 with 12+ comments confirming bug persistence.

3.2 Problem Description

When setting up a histogram with a logarithmic scale on “auto”, the Y-axis maximum limit gets stuck at 1k (100 in code), regardless of the actual data values.

Context

Workaround Available: Setting explicit Y-axis maximum values manually works correctly. The issue is specifically with the “auto” limit calculation for log scales.

3.3 Critical Finding: Hardcoded Y-Axis

Key Finding

Found in public/app/plugins/panel/histogram/Histogram.tsx line 152: The histogram panel hardcodes the Y-axis scale distribution to ScaleDistribution.Linear.

```
builder.addScale({
  scaleKey: 'y', // counts
  isTime: false,
  distribution: ScaleDistribution.Linear, // HARDCODED!
  orientation: ScaleOrientation.Vertical,
  direction: ScaleDirection.Up,
  softMin: 0,
});
```

3.4 X-Axis vs Y-Axis Comparison

Aspect	X-Axis	Y-Axis	Status
Scale Distribution	Configurable	Hardcoded Linear	×
Log Scale Support	Yes	No	×
Range Function	Custom for log	Default linear	×
Auto-scaling	Works	Broken for log	×

Important

Note: The X-axis properly supports log scale based on bucket size differences, but the Y-axis does not share this capability.

4 Engineering Quality Assessment

This section evaluates the investigation artifacts against professional engineering documentation standards.

4.1 Visual Design Elements

Element	Implementation
Visual Hierarchy	Color-coded headings, consistent spacing, clear section breaks
Typography	Professional Times/Helvetica font family, proper weight hierarchy
Tables	Clean lines, light backgrounds, professional borders
Callout Boxes	Three distinct types with color coding
Code Presentation	Syntax highlighted with pastie style
Branding	KDE color scheme consistently applied

5 Conclusions

5.1 Investigation Success

Key Finding

- **Issue #116074:** Root cause identified in label expansion pipeline
- **Issue #13027:** Root cause identified in hardcoded Y-axis scale

5.2 Next Steps

Per the Five Core Principles, implementation requires human authorization:

1. Review investigation findings
2. Authorize implementation plans
3. Proceed with code changes
4. Execute test verification
5. Submit pull requests to Grafana

Appendix: Artifact Locations

Artifact	File Path
Investigation Report 1	INV-LAB039-001.md
Implementation Plan 1	INV-LAB039-001-PLAN.md
Investigation Report 2	INV-LAB039-002.md
Implementation Plan 2	INV-LAB039-002-PLAN.md
LaTeX Source	LAB-040/engineering-report.tex

Document Status: Investigation Complete
Awaiting Authorization for Implementation

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